Exploration–Exploitation Algorithms for Credit Scoring

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Abstract

We study three multi-armed bandit algorithms—ε-greedy, Upper Confidence Bound (UCB1), and Thompson Sampling for Bernoulli rewards—in a credit scoring context where each arm is a repeat-borrower segment and rewards encode repayment success. Over T=5000 borrower interactions and 50 Monte Carlo simulations, UCB1 with exploration coefficient c=0.25 achieved the highest mean total reward (3634.3 successful repayments), while Thompson Sampling converged fastest (t^*=6) and concentrated most strongly on the true best segment in the final 10% of pulls (tail best-arm rate 0.84). We compare exploration efficiency, exploitation effectiveness, convergence, and suitability for adaptive credit scoring using cumulative reward, average reward, and arm-selection frequency plots.

1 Introduction

Credit institutions increasingly score *repeat borrowers* using past repayment behavior. When several borrower segments, policy tiers, or applicant categories are available, the lender must balance **exploration** (trying less-understood groups to learn their repayment rates) and **exploitation** (allocating decisions to groups believed most creditworthy). This assignment frames that problem as a **multi-armed bandit**: each arm is a segment; each interaction yields a Bernoulli reward (1 = repayment, 0 = default).

We implement and evaluate:

1. ε-greedy exploration,
2. UCB1 with tunable exploration parameter c,
3. Thompson Sampling with Beta–Bernoulli posteriors.

2 Problem Setup

2.1 Arms and rewards

We simulate K=6 repeat-borrower segments with true repayment probabilities

$$\mathbf{p} = (0.58,\ 0.61,\ 0.65,\ 0.69,\ 0.72,\ 0.74).$$

Arm 5 is the most creditworthy ($p^* = 0.74$). At each step $t \in \{1, \ldots, T\}$, an algorithm selects an arm a_t, observes $r_t \sim \text{Bernoulli}(p_{a_t})$, and updates estimates used as **creditworthiness scores** for ranking segments.

2.2 Metrics

Over S=50 independent simulations we report:

- **Mean total reward**: $\bar{R} = \frac{1}{S} \sum_s \sum_{t=1}^{T} r_t^{(s)}$.

- **Mean average reward**: $\bar{R}/T$.
- **Mean regret**: $Tp^* - \bar{R}$ (oracle always pulls the best arm).
- **Tail best-arm rate**: fraction of pulls of arm 5 in the last 10% of steps.
- **Convergence t^***: first step where running average reward $\geq 0.95\, p^*$ (lower is faster).
- **Mean final ranking**: arms ordered by learned value estimates (descending).

Experiments use T=5000, S=50, seed 7 (see `run_experiments.py`).

3 Algorithms

3.1 Epsilon-greedy

With probability ε, select a uniform random arm; otherwise exploit the arm with highest sample-mean reward $\hat{\mu}_a$. Incremental updates: $\hat{\mu}_a \leftarrow \hat{\mu}_a + (r - \hat{\mu}_a)/n_a$.

3.2 UCB1

After an initial pull of each arm, select

$$a_t = \arg\max_a \left(\hat{\mu}_a + c\sqrt{\frac{2\ln t}{n_a}} \right),$$

where $c \geq 0$ controls optimism toward under-sampled arms.

3.3 Thompson Sampling (Bernoulli)

Maintain $\mathrm{Beta}(\alpha_a, \beta_a)$ posteriors per arm. Each step sample $\tilde{\theta}_a \sim \mathrm{Beta}(\alpha_a, \beta_a)$, play $\arg\max_a \tilde{\theta}_a$, then update $\alpha_a \mathrel{+}= r$, $\beta_a \mathrel{+}= 1 - r$.

4 Experimental Results

4.1 Epsilon-greedy sweep

Table 1 summarizes the effect of ε on exploration versus exploitation.

Table 1: Epsilon-greedy sweep (T=5000, S=50).

Setting	Total reward	Avg reward	Regret	Tail best-arm	t^*
$\varepsilon = 0.01$	3575.7	0.7151	124.3	0.555	268
$\varepsilon = 0.05$	**3622.7**	**0.7245**	**77.3**	**0.708**	96
$\varepsilon = 0.10$	3614.1	0.7228	85.9	0.690	239
$\varepsilon = 0.20$	3587.0	0.7174	113.0	0.640	91

Observation. ε=0.05 dominates the sweep: 47.0 more successful repayments than ε=0.01 and 35.7 more than ε=0.20, with the lowest regret (77.3). Very small ε=0.01 under-explores (regret 124.3, slow t^*=268). Larger ε values waste pulls on random segments after the best arm is identifiable, lowering total reward.

4.2 UCB sweep

Table 2 shows how the exploration coefficient c affects performance.

Observation. Smaller c=0.25 yields the best total reward and lowest regret. As c increases, UCB over-explores: total reward drops by 205.3 from c=0.25 to c=2.00, and tail best-arm rate falls from 0.718 to 0.300, indicating weaker late-stage focus on arm 5.

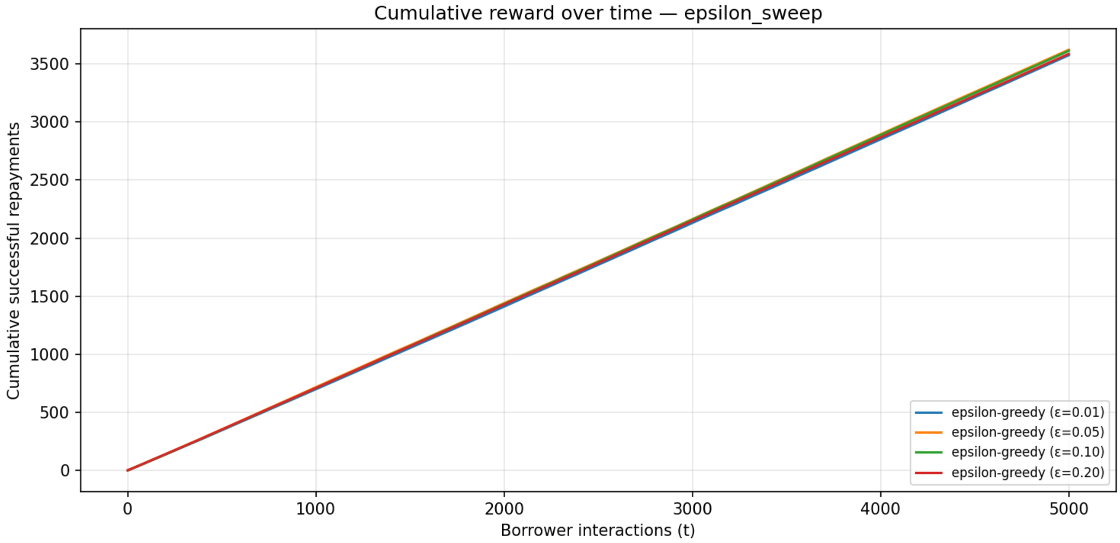

Figure 1: Cumulative reward over time—epsilon-greedy sweep.

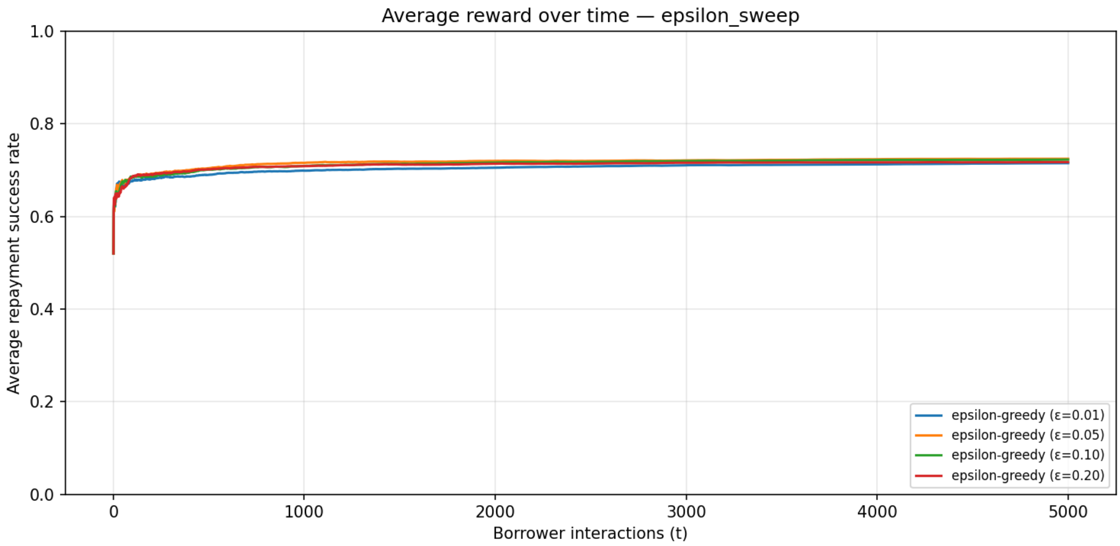

Figure 2: Average repayment success rate over time—epsilon-greedy sweep.

Table 2: UCB1 sweep (T=5000, S=50).

Setting	Total reward	Avg reward	Regret	Tail best-arm	t^*
$c = 0.25$	**3634.3**	**0.7269**	**65.7**	**0.718**	16
$c = 0.50$	3594.9	0.7190	105.1	0.694	51
$c = 1.00$	3503.5	0.7007	196.5	0.451	182
$c = 2.00$	3429.0	0.6858	271.0	0.300	23

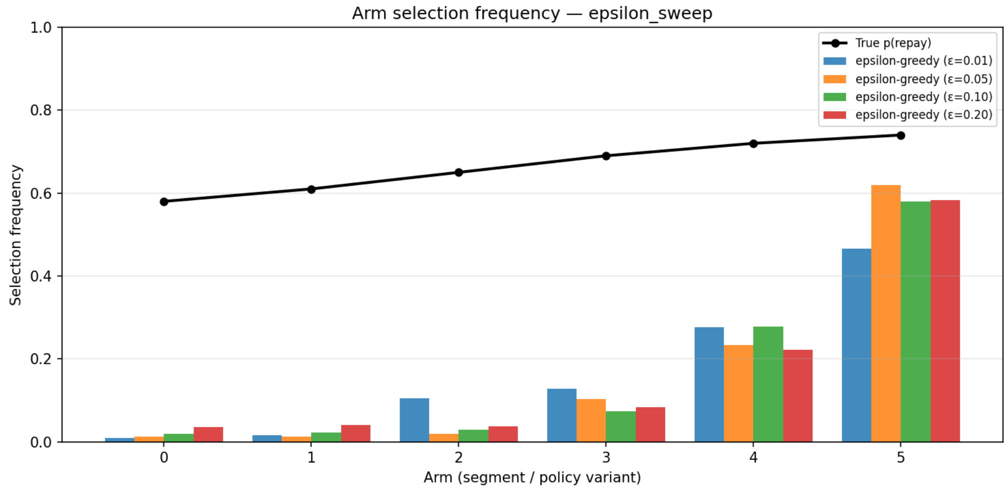

Figure 3: Arm selection frequencies vs. true repayment probabilities—epsilon-greedy sweep.

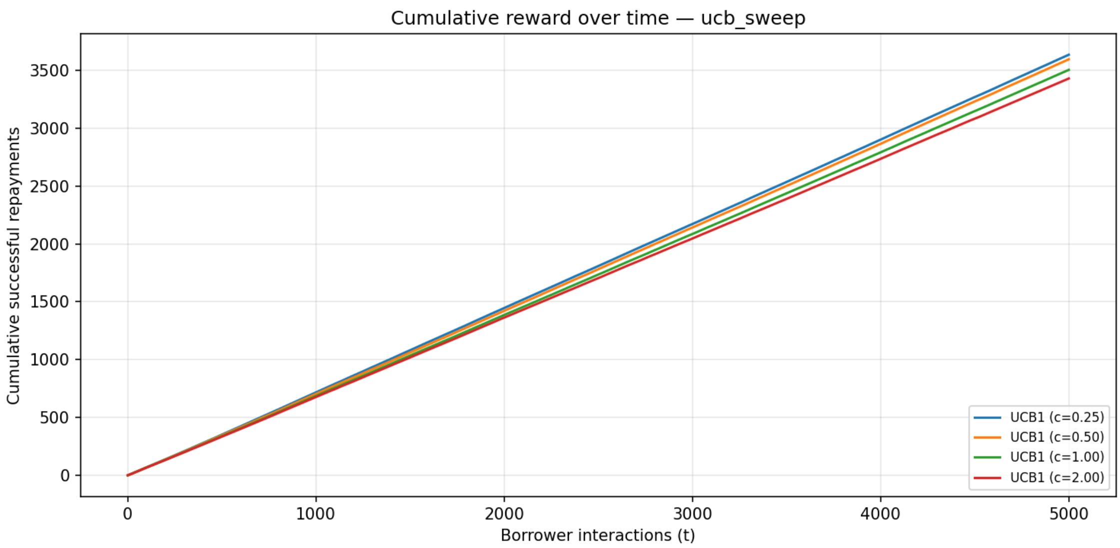

Figure 4: Cumulative reward over time—UCB sweep.

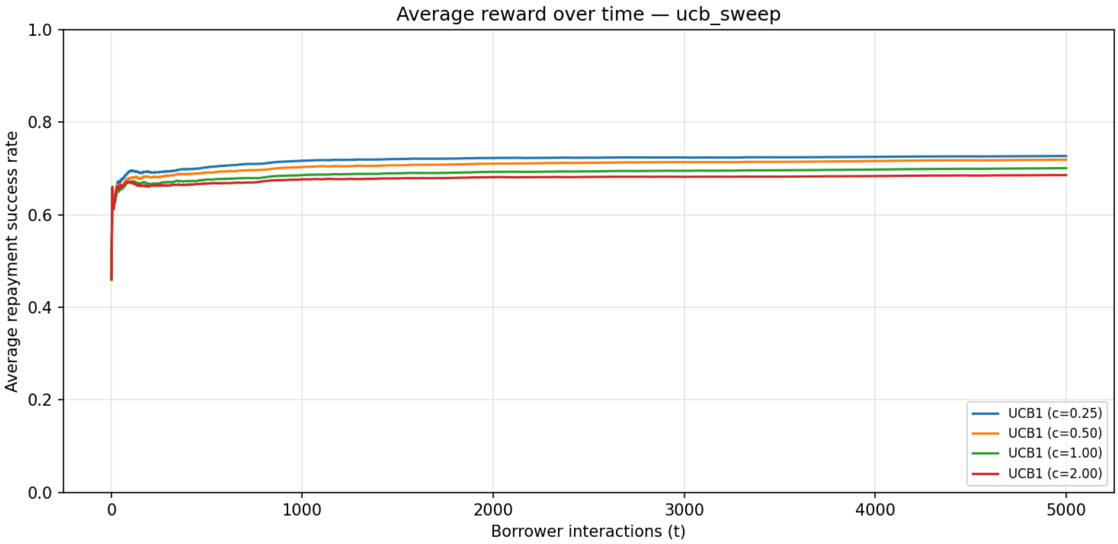

Figure 5: Average repayment success rate over time—UCB sweep.

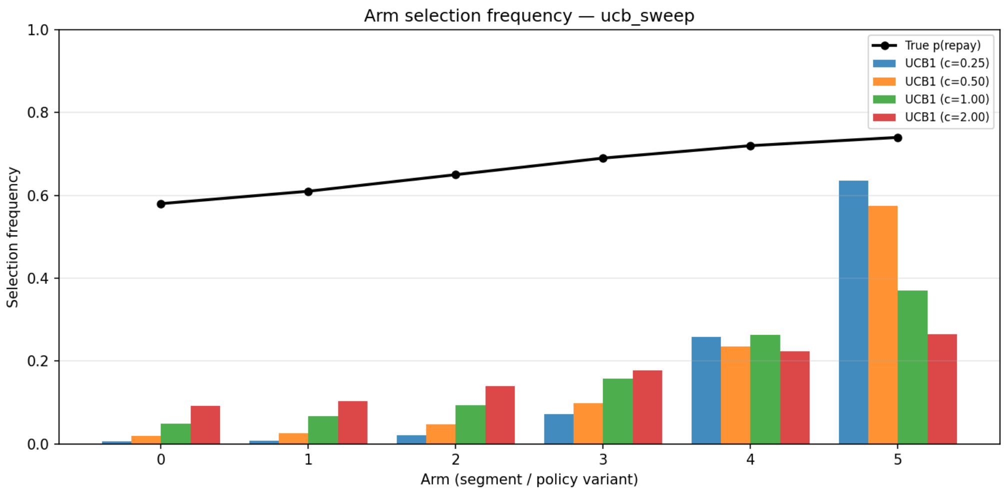

Figure 6: Arm selection frequencies—UCB sweep.

4.3 Final comparison

Table 3 compares the best hyperparameters from each sweep against Thompson Sampling.

Table 3: Final comparison: best ε-greedy, best UCB1, Thompson Sampling.

Method	Total reward	Avg reward	Regret	Tail best-arm	t^*	Ranking
ε-greedy (ε=0.05)	3622.7	0.7245	77.3	0.708	96	5>4>3>2>1>0
UCB1 (c=0.25)	**3634.3**	**0.7269**	**65.7**	0.718	16	5>4>3>2>1>0
Thompson Sampling	3626.4	0.7253	73.6	**0.840**	**6**	5>4>3>2>1>0

The oracle benchmark is $Tp^* = 5000 \times 0.74 = 3700$ repayments. UCB1 (c=0.25) comes closest (65.7 regret). Thompson Sampling ranks second on total reward (3626.4) but achieves the fastest convergence (t^*=6) and the strongest late exploitation of the best arm (tail rate 0.84).

Figure 7: Cumulative reward—final comparison.

5 Discussion and Analysis

5.1 Total reward earned

Across T=5000 interactions, UCB1 with c=0.25 earned the most repayments (3634.3, average rate 0.7269), outperforming the best ε-greedy setting by 11.6 repayments and Thompson Sampling by 7.9. All three algorithms improved cumulative reward over time (Figures 1–7), approaching but not reaching the oracle rate 0.74 because exploration allocates some pulls to suboptimal segments.

5.2 Exploration efficiency

Regret quantifies wasted repayments relative to always selecting arm 5. UCB (c=0.25) achieved the lowest regret among final methods (65.7), versus 77.3 for ε-greedy and 73.6 for Thompson Sampling. Within the UCB sweep, increasing c from 0.25 to 2.00 more than quadrupled regret (65.7 $\rightarrow$ 271.0), showing that unstructured optimism can be as costly as excessive random

Figure 8: Average reward—final comparison.

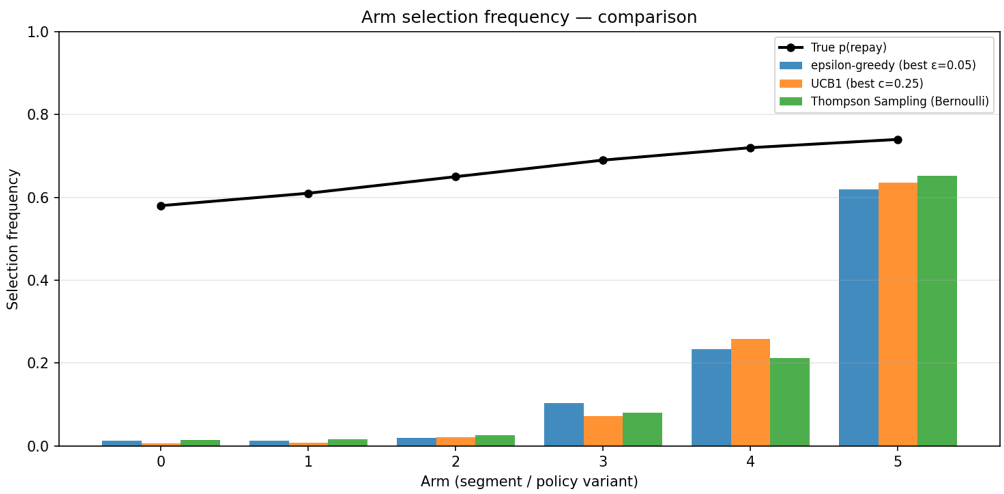

Figure 9: Arm selection frequencies—final comparison. Black overlay: true p_a.

exploration in ε-greedy (ε=0.20, regret 113.0). UCB and Thompson Sampling explore *where uncertainty is high*, whereas ε-greedy explores uniformly at rate ε.

5.3 Exploitation effectiveness

Tail best-arm rate measures how often the algorithm selects the true best segment once learning has progressed. Thompson Sampling leads (0.840), followed by UCB (0.718) and ε-greedy (0.708). The arm-frequency plots (Figures 3, 6, 9) show increasing concentration on arm 5 for well-tuned settings; overtuned UCB (c=2.00, tail rate 0.300) fails to exploit arm 5 consistently. ε-greedy continues random exploration even after convergence, capping exploitation relative to Thompson Sampling.

5.4 Convergence speed

Thompson Sampling reached 95% of p^* fastest (t^*=6), then UCB (16), then ε-greedy (96). Average-reward curves (Figures 8 and sweep plots) rise steeply for Thompson Sampling and UCB early in the horizon. ε=0.01 was particularly slow (t^*=268) because it rarely probes alternative segments. Moderate ε=0.05 balanced speed and final performance within the epsilon sweep.

5.5 Identifying the most creditworthy repeat borrowers

All three final methods recovered the correct mean learned ranking ($5 > 4 > 3 > 2 > 1 > 0$), matching the true repayment ordering. Thompson Sampling's posterior means provide interpretable segment-level repayment beliefs. High tail best-arm rates and alignment of selection frequency with true p_a in Figure 9 indicate successful identification of the most creditworthy repeat-borrower segment.

5.6 Suitability for adaptive credit scoring

- **ε-greedy** is a simple baseline for adaptive reapplication scoring but requires tuning ε. Our results show a 47-repayment gap between the best and worst ε in the sweep.
- **UCB1** is well suited when the system must learn systematically from limited segment data: best total reward and low regret with c=0.25, with deterministic decisions given history.
- **Thompson Sampling** is attractive when the lender wants calibrated uncertainty (Beta posteriors per segment), fast early learning (t^*=6), and strong late focus on the best segment (tail rate 0.84), at a small cost in total reward versus UCB in this simulation.

Overall, exploration–exploitation bandits support repeat-borrower credit scoring by continuously updating segment creditworthiness from repayment outcomes. For this synthetic scenario, **UCB1** maximized repayments, while **Thompson Sampling** offered the fastest convergence and strongest identification of the best arm in the final phase.

6 Implementation

Source code: `bandits.py` (algorithms), `run_experiments.py` (experiments and plots), `svg_plots.py` (visualization). Run this command to get results:

```
python run_experiments.py --steps 5000 --sims 50 --seed 7 --outdir outputs
```

References

- Auer, Cesa-Bianchi, and Fischer (2002). Finite-time Analysis of the Multiarmed Bandit Problem.
- Thompson (1933). On the Likelihood that One Unknown Probability Exceeds Another.
- Sutton and Barto (2018). *Reinforcement Learning: An Introduction* (bandits chapter).